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Smart Docklands Smart Bin Innovation Services – Project Proposal

GitHub Repo: https://github.com/Ikz333/Smart-Bin-App.git Master Branch

Theme:

Smart Bin Innovation Services

Application Overview

This project focuses on the Smart Bin Innovation Services theme from the Smart Docklands initiative. The goal of the application is to simulate a smart waste management system that could be used in Dublin's Smart Docklands area. The system allows users to manage smart bins digitally through a graphical user interface built using Java Swing in NetBeans.

The application allows users to add, search, update, and delete smart bin records. Each bin stores information such as the bin ID, location, zone, bin type, fill level, battery level, and status. These bins are stored as objects in an ArrayList, which allows the system to manage multiple bins efficiently and perform CRUD (Create, Read, Update, Delete) operations.

The application also uses several Abstract Data Types (ADTs) to organise data in different ways. In addition to the ArrayList, the system includes a Queue, Stack, and Singly Linked List. The Queue is used to simulate the order in which bins should be collected using a first-in-first-out structure. The Stack is used to store actions performed in the system, which allows actions to be removed in a last-in-first-out order. A Singly Linked List is also included to store notes or alerts related to the system, such as maintenance messages. The linked list panel allows users to add, search, update, delete, and display items, demonstrating CRUD operations on another data structure.

Smart bins are an important part of modern smart city infrastructure. These bins can monitor how full they are and notify waste collection services when they need to be emptied. The application simulates this by allowing the user to enter the fill level of a bin and determine its collection priority.

Problem Being Solved

Traditional waste collection systems usually follow fixed schedules. This can cause problems such as bins overflowing before they are collected or trucks collecting bins that are not yet full. These issues can lead to inefficiency, increased costs, and negative environmental impacts.

The Smart Bin application demonstrates how smart technology can improve waste management by tracking bin information and identifying which bins need to be collected

first. Each bin stores its fill level, which represents how full the bin is. The system uses this information to determine a collection priority. For example, organic bins may require quicker collection compared to general waste bins.

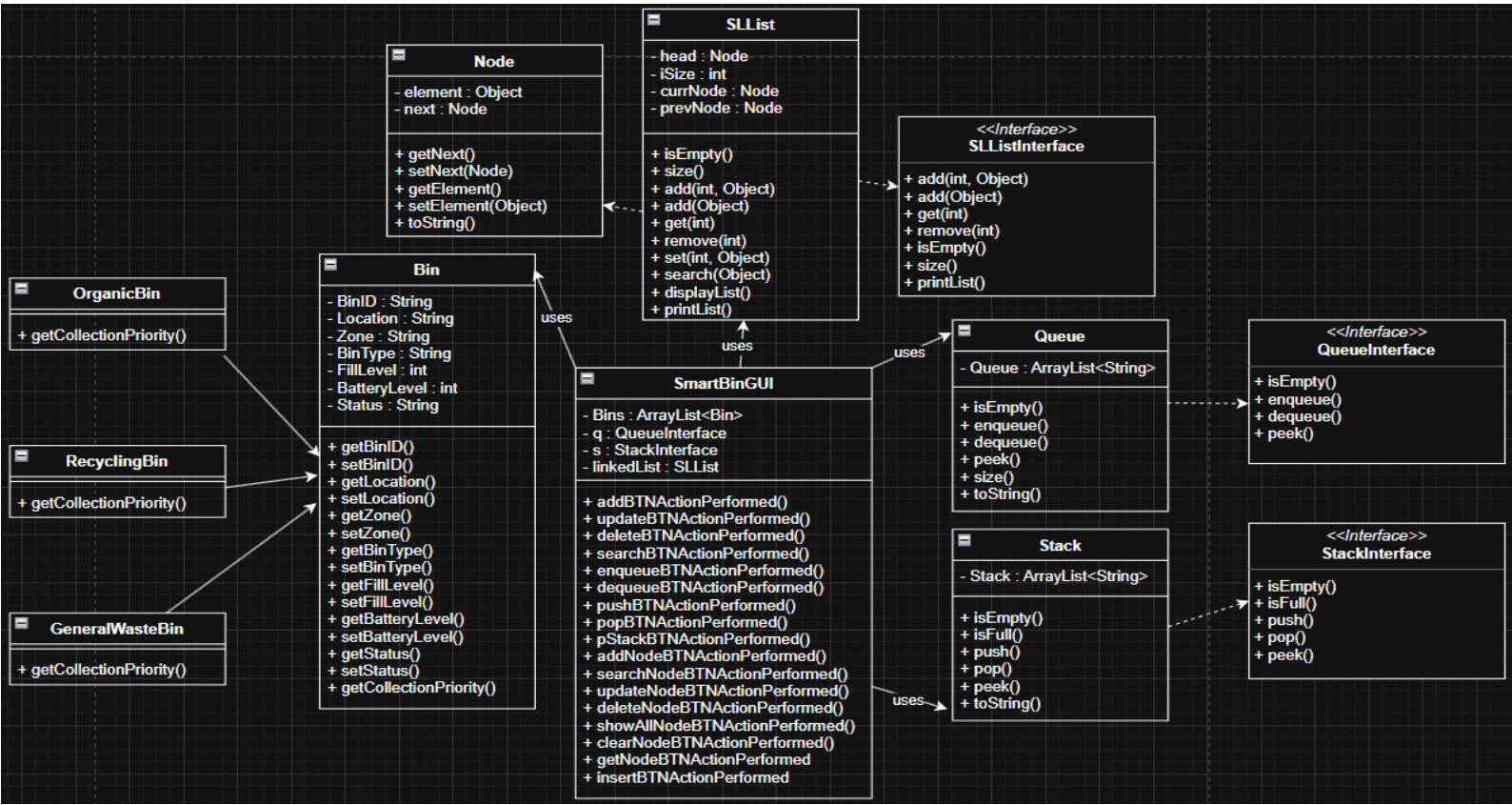
By using this type of system, waste collection teams could prioritise bins that are close to full instead of following a fixed schedule. This would help improve efficiency, reduce unnecessary trips, and keep public areas cleaner.

How the Application Fits the Theme

The Smart Docklands initiative focuses on testing innovative technologies that can improve services in Dublin's Docklands area. One of its goals is to improve city services using digital solutions and data-driven systems.

The Smart Bin Innovation Services application supports this goal by demonstrating how technology can improve waste management. The system models a simplified smart bin network where bin data can be stored, monitored, and prioritised for collection.

The application also demonstrates important programming concepts such as object-oriented programming, inheritance, polymorphism, graphical user interfaces, and data structures. Bins are represented using a base Bin class and subclasses such as GeneralWasteBin, RecyclingBin, and OrganicBin, which demonstrate inheritance and polymorphism.

Overall, this project shows how a smart waste management system could operate in a smart city environment while applying core programming concepts and data structures.

Class Diagram